



**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

# **LESSON PLAN**

For

## **CompTIA Certified Network+ Training (N10-009)**

TGS Ref No: TGS-2025054472

Conducted by

**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

**Version 1.0**

## DOCUMENT VERSION CONTROL RECORD

| Version Number | Effective Date of Release | Summary of Included Changes                                                                           | Author         |
|----------------|---------------------------|-------------------------------------------------------------------------------------------------------|----------------|
| 1.0            | 2 July 2026               | Initial release —<br>Network+ (N10-009) 5-day lesson plan aligned to the 36 labs and the course deck. | Dr. Alfred Ang |

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## Course Information

|                             |                                                                   |
|-----------------------------|-------------------------------------------------------------------|
| <b>Course Title</b>         | CompTIA Certified Network+ Training (N10-009)                     |
| <b>WSQ Course Reference</b> | TGS-2025054472                                                    |
| <b>Training Provider</b>    | Tertiary Infotech Academy Pte Ltd (UEN 201200696W)                |
| <b>Duration</b>             | 5 days · 8 training hours per day (40 hours)                      |
| <b>Daily Timing</b>         | 9:00 am – 6:00 pm (1-hour lunch; tea breaks within training time) |
| <b>Mode</b>                 | Instructor-led, hands-on labs on the Killercoda Ubuntu Playground |
| <b>Trainer</b>              | Dr. Alfred Ang                                                    |

## Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Explain networking concepts — the OSI model, IPv4/IPv6 addressing and subnetting, ports and protocols, DNS and DHCP, network topologies, architectures and cloud connectivity.
- LO2: Implement networks — static and dynamic routing, 802.1Q VLANs, trunking, link aggregation, NAT/PAT, wireless, and core DHCP and DNS services.
- LO3: Operate and maintain networks — SNMP monitoring, flow analysis, centralized logging and SIEM, time synchronization, configuration backup and performance baselining.
- LO4: Secure networks — firewalls, ACLs, IDS/IPS, TLS certificates, SSH hardening, VPNs, port scanning and access control.
- LO5: Troubleshoot networks — apply the CompTIA seven-step methodology using layer-by-layer diagnostics, connectivity tools and packet capture.

## Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — hands-on networking tasks, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 5 from 4:00 pm (WA 4:00–5:00 pm, PP 5:00–6:00 pm).
- A minimum of 75% attendance is required to be eligible for assessment and funding.
- Courseware and the assessment are on the LMS (<https://lms-tms.tertiaryinfotech.com/>).

## Course Schedule

The Slides column gives the matching page range in the course deck so the trainer can run each session directly from the slides.

## Day 1 — Networking Concepts

| Time              | Duration | Topic / Activity                                                                                                                                                                     | Slides       |
|-------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 9:00–9:30         | 30 min   | Welcome, course introduction, learning outcomes, ground rules and mandatory digital attendance (AM)                                                                                  | —            |
| <b>9:30–10:30</b> | 60 min   | <b>Domain 1.0 — Networking Concepts: OSI model, encapsulation, IPv4/IPv6 addressing, ports &amp; protocols, DNS, DHCP, topologies, architectures and cloud (concepts + diagrams)</b> | Slides 22–25 |
| 10:30–10:45       | 15 min   | Tea break                                                                                                                                                                            | —            |
| 10:45–13:00       | 135 min  | Hands-on: Lab 1: OSI Model Packet Inspection; Lab 2: IPv4 Subnetting and CIDR; Lab 3: IPv6 Addressing and Link-Local; Lab 4: Common Ports and Protocols                              | Slides 26–61 |
| 13:00–14:00       | 60 min   | Lunch break                                                                                                                                                                          | —            |
| 14:00–15:30       | 90 min   | Hands-on: Lab 5: DNS Record Types; Lab 6: DHCP Lease Capture (DORA)                                                                                                                  | Slides 62–82 |
| 15:30–15:45       | 15 min   | Tea break                                                                                                                                                                            | —            |
| 15:45–17:45       | 120 min  | Hands-on: Lab 7: Cloud VPC with Network Namespaces. Domain 1 review.                                                                                                                 | Slides 83–96 |
| 17:45–18:00       | 15 min   | Day 1 recap, Q&A and PM digital attendance                                                                                                                                           | —            |

Total training time: 480 minutes (8 hours).

## Day 2 — Network Implementation

| Time              | Duration | Topic / Activity                                                                                                                               | Slides       |
|-------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 9:00–9:15         | 15 min   | Day 1 recap and mandatory digital attendance (AM)                                                                                              | —            |
| <b>9:15–10:30</b> | 75 min   | <b>Domain 2.0 — Network Implementation: routing, VLANs, trunking, link aggregation, NAT/PAT, wireless, DHCP &amp; DNS services (concepts +</b> | Slides 93–96 |

|             |         |                                                                                                                                                         |                |
|-------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
|             |         | <b>diagrams)</b>                                                                                                                                        |                |
| 10:30–10:45 | 15 min  | Tea break                                                                                                                                               | —              |
| 10:45–13:00 | 135 min | Hands-on: Lab 8: Static Routing Between Subnets; Lab 9: 802.1Q VLAN Tagging; Lab 10: Link Aggregation (LACP Bond); Lab 11: Wireless Concepts and Survey | Slides 97–131  |
| 13:00–14:00 | 60 min  | Lunch break                                                                                                                                             | —              |
| 14:00–15:30 | 90 min  | Hands-on: Lab 12: NAT and PAT with iptables; Lab 13: DHCP Server (isc-dhcp-server)                                                                      | Slides 132–150 |
| 15:30–15:45 | 15 min  | Tea break                                                                                                                                               | —              |
| 15:45–17:45 | 120 min | Hands-on: Lab 14: Authoritative DNS with BIND9. Domain 2 review.                                                                                        | Slides 151–164 |
| 17:45–18:00 | 15 min  | Day 2 recap, Q&A and PM digital attendance                                                                                                              | —              |

Total training time: 480 minutes (8 hours).

### Day 3 — Network Operations

| Time              | Duration | Topic / Activity                                                                                                                                                     | Slides         |
|-------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 9:00–9:15         | 15 min   | Day 2 recap and mandatory digital attendance (AM)                                                                                                                    | —              |
| <b>9:15–10:30</b> | 75 min   | <b>Domain 3.0 — Network Operations: SNMP, NetFlow, syslog &amp; SIEM, NTP, configuration backup, performance baselines &amp; documentation (concepts + diagrams)</b> | Slides 161–164 |
| 10:30–10:45       | 15 min   | Tea break                                                                                                                                                            | —              |
| 10:45–13:00       | 135 min  | Hands-on: Lab 15: SNMP Monitoring; Lab 16: NetFlow Analysis with nfdump; Lab 17: Centralised Logging (rsyslog)                                                       | Slides 165–190 |
| 13:00–14:00       | 60 min   | Lunch break                                                                                                                                                          | —              |
| 14:00–15:30       | 90 min   | Hands-on: Lab 18: Time Sync with Chrony; Lab 19: Configuration Backup with Git                                                                                       | Slides 191–207 |
| 15:30–15:45       | 15 min   | Tea break                                                                                                                                                            | —              |
| 15:45–17:45       | 120 min  | Hands-on: Lab 20: Performance Baseline                                                                                                                               | Slides 208–222 |

|             |        |                                            |   |
|-------------|--------|--------------------------------------------|---|
|             |        | with iperf3. Domain 3 review.              |   |
| 17:45–18:00 | 15 min | Day 3 recap, Q&A and PM digital attendance | — |

Total training time: 480 minutes (8 hours).

## Day 4 — Network Security

| Time              | Duration | Topic / Activity                                                                                                                                                      | Slides         |
|-------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 9:00–9:15         | 15 min   | Day 3 recap and mandatory digital attendance (AM)                                                                                                                     | —              |
| <b>9:15–10:30</b> | 75 min   | <b>Domain 4.0 — Network Security: CIA triad, firewalls, ACLs, screened subnets, IDS/IPS, TLS, SSH, VPNs, port scanning &amp; access control (concepts + diagrams)</b> | Slides 219–222 |
| 10:30–10:45       | 15 min   | Tea break                                                                                                                                                             | —              |
| 10:45–13:00       | 135 min  | Hands-on: Lab 21: Host Firewall with UFW; Lab 22: TLS Certificate with OpenSSL + Nginx; Lab 23: SSH Hardening with Key Auth; Lab 24: Site-to-Site VPN with WireGuard  | Slides 223–259 |
| 13:00–14:00       | 60 min   | Lunch break                                                                                                                                                           | —              |
| 14:00–15:30       | 90 min   | Hands-on: Lab 25: Port Scanning and Hardening; Lab 26: Suricata IDS                                                                                                   | Slides 260–278 |
| 15:30–15:45       | 15 min   | Tea break                                                                                                                                                             | —              |
| 15:45–17:45       | 120 min  | Hands-on: Lab 27: ACL with iptables. Domain 4 review.                                                                                                                 | Slides 279–293 |
| 17:45–18:00       | 15 min   | Day 4 recap, Q&A and PM digital attendance                                                                                                                            | —              |

Total training time: 480 minutes (8 hours).

## Day 5 — Network Troubleshooting & Assessment

| Time              | Duration | Topic / Activity                                                                                                                    | Slides         |
|-------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 9:00–9:15         | 15 min   | Day 4 recap and mandatory digital attendance (AM)                                                                                   | —              |
| <b>9:15–10:15</b> | 60 min   | <b>Domain 5.0 — Network Troubleshooting: the seven-step methodology, layer 1/2 diagnostics, connectivity tools, DNS/routing/MTU</b> | Slides 290–293 |

|             |         |                                                                                                                                                                               |                |
|-------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
|             |         | <b>issues and packet capture (concepts + diagrams)</b>                                                                                                                        |                |
| 10:15–10:30 | 15 min  | Tea break                                                                                                                                                                     | —              |
| 10:30–13:00 | 150 min | Hands-on: Lab 28: Layer 1/2 Diagnostics; Lab 29: ping, traceroute, mtr; Lab 30: DNS Troubleshooting; Lab 31: ARP and Neighbour Table; Lab 32: Packet Capture (tcpdump/tshark) | Slides 294–340 |
| 13:00–14:00 | 60 min  | Lunch break                                                                                                                                                                   | —              |
| 14:00–15:30 | 90 min  | Hands-on: Lab 33: MTU and Path-MTU Discovery; Lab 34: Routing Issues; Lab 35: Network Emulation with tc; Lab 36: Service Reachability Triage                                  | Slides 341–384 |
| 15:30–15:45 | 15 min  | Tea break                                                                                                                                                                     | —              |
| 15:45–16:00 | 15 min  | <b>Briefing for Assessment</b>                                                                                                                                                | —              |
| 16:00–17:00 | 60 min  | <b>Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book</b>                                                                                              | —              |
| 17:00–18:00 | 60 min  | <b>Practical Performance (PP) — hands-on networking tasks, 1 hour, open book. PM digital attendance</b>                                                                       | —              |

Total training time: 480 minutes (8 hours).

## Topic-by-Topic Breakdown

Each domain and lab below shows its starting slide in the course deck.

### Domain 1.0 — Networking Concepts (23%) • Slides 22–96

OSI Model · IPv4/IPv6 Addressing · Ports & Protocols · DNS & DHCP · Topologies & Cloud

#### Lab 1 — OSI Model Packet Inspection • Slide 26

**Objective:** Inspect packets and map headers to OSI layers.

**Activity:** The learner captures live traffic with tcpdump and maps each protocol header to its OSI layer.



### Lab 2 — IPv4 Subnetting and CIDR • Slide 35

**Objective:** Subnet a network into cidr blocks and verify boundaries.

**Activity:** The learner splits a /24 network into four /26 subnets, verifies the math in the TIS IP Calculator, and assigns a CIDR address on Linux.

### Lab 3 — IPv6 Addressing and Link-Local • Slide 44

**Objective:** Plan ipv6 prefixes and distinguish link-local from global addresses.

**Activity:** The learner plans an IPv6 prefix in the TIS IP Calculator, adds a global address, observes the auto-generated link-local address, and pings across both.

### Lab 4 — Common Ports and Protocols • Slide 54

**Objective:** Map common protocol names to their well-known ports.

**Activity:** The learner enumerates listening services and scans a live target to connect protocol names to their well-known port numbers.

### Lab 5 — DNS Record Types • Slide 62

**Objective:** Query each dns record type on the exam blueprint.

**Activity:** The learner uses dig to query every DNS record type in the N10-009 objectives: A, AAAA, MX, TXT, NS, CNAME, PTR, SOA, and SRV.

### Lab 6 — DHCP Lease Capture (DORA) • Slide 75

**Objective:** Capture the four-step dhcp dora exchange.

**Activity:** The learner releases an existing DHCP lease and captures the four DORA messages (Discover, Offer, Request, Acknowledge) live with tcpdump.

### Lab 7 — Cloud VPC with Network Namespaces • Slide 83

**Objective:** Simulate a cloud vpc using linux network namespaces.

**Activity:** The learner builds a VPC with two subnets and a bridge using Linux network namespaces and veth pairs, then tests connectivity between the two virtual machines.

## Domain 2.0 — Network Implementation (20%) • Slides 93–164

Routing · VLANs & Trunking · Link Aggregation · NAT/PAT · Wireless · DHCP & DNS Services

### Lab 8 — Static Routing Between Subnets • Slide 97

**Objective:** Configure static routes between subnets.

**Activity:** The learner builds a three-node subnet A - router - subnet B topology and adds the static routes that let the two subnets communicate.

### Lab 9 — 802.1Q VLAN Tagging • Slide 107

**Objective:** Create 802.1q vlan sub-interfaces.

**Activity:** The learner loads the 802.1Q kernel module and creates two tagged VLAN sub-interfaces on Linux to separate broadcast domains.

**Lab 10 — Link Aggregation (LACP Bond) · Slide 116**

**Objective:** Bundle nics into an lacp bond.

**Activity:** The learner installs ifenslave, loads the bonding driver in 802.3ad mode and creates an LACP bond interface with an enslaved NIC.

**Lab 11 — Wireless Concepts and Survey · Slide 124**

**Objective:** Survey wireless bands, channels and security.

**Activity:** The learner surveys wireless bands and channels with iw and compares 2.4/5/6 GHz behaviour and WPA security modes.

**Lab 12 — NAT and PAT with iptables · Slide 132**

**Objective:** Translate private ips with nat and pat.

**Activity:** The learner configures an iptables MASQUERADE rule so hosts on a private subnet share the host public IP via port address translation.

**Lab 13 — DHCP Server (isc-dhcp-server) · Slide 141**

**Objective:** Configure a dhcp scope and lease server.

**Activity:** The learner installs isc-dhcp-server and configures a DHCP scope on a dummy interface to hand out leases on a private subnet.

**Lab 14 — Authoritative DNS with BIND9 · Slide 151**

**Objective:** Serve an authoritative dns zone.

**Activity:** The learner runs a BIND9 server hosting the authoritative zone lab.local and queries its records with dig.

**Domain 3.0 — Network Operations (19%) · Slides 161-222**

SNMP · NetFlow · Syslog & SIEM · NTP · Config Backup · Performance Baselines

**Lab 15 — SNMP Monitoring · Slide 165**

**Objective:** Monitor devices with snmp.

**Activity:** The learner installs an SNMP agent, configures a read-only community string, and walks the MIB to read device metrics.

**Lab 16 — NetFlow Analysis with nfdump · Slide 173**

**Objective:** Export and analyse network flows.

**Activity:** The learner runs a NetFlow exporter and collector on one VM and analyses the captured flows with nfdump.

#### Lab 17 — Centralised Logging (rsyslog) • Slide 182

**Objective:** Centralise logs with syslog forwarding.

**Activity:** The learner configures rsyslog to both send and receive on loopback, proving the log-forwarding pipeline.

#### Lab 18 — Time Sync with Chrony • Slide 191

**Objective:** Synchronise clocks with ntp.

**Activity:** The learner installs chrony, inspects its time sources and tracking state, and forces an immediate sync.

#### Lab 19 — Configuration Backup with Git • Slide 199

**Objective:** Version-control network configuration.

**Activity:** The learner places /etc under Git version control to gain change history, audit trail, and rollback.

#### Lab 20 — Performance Baseline with iperf3 • Slide 208

**Objective:** Baseline throughput, jitter, and loss.

**Activity:** The learner runs an iperf3 server and client to measure TCP throughput, parallel streams, UDP jitter/loss, and reverse-mode traffic.

### Domain 4.0 — Network Security (14%) • Slides 219–293

CIA Triad • Firewalls & ACLs • IDS/IPS • TLS • SSH • VPNs • NAC

#### Lab 21 — Host Firewall with UFW • Slide 223

**Objective:** Configure a host firewall.

**Activity:** The learner configures a default-deny host firewall with UFW, permitting only required inbound services.

#### Lab 22 — TLS Certificate with OpenSSL + Nginx • Slide 233

**Objective:** Serve https with a tls certificate.

**Activity:** The learner generates a self-signed X.509 certificate, serves it from nginx, and verifies the TLS handshake.

#### Lab 23 — SSH Hardening with Key Auth • Slide 242

**Objective:** Harden ssh with key authentication.

**Activity:** The learner creates an Ed25519 key pair, installs the public key, and disables SSH password authentication.

#### Lab 24 — Site-to-Site VPN with WireGuard • Slide 250

**Objective:** Build a site-to-site vpn tunnel.

**Activity:** The learner builds a WireGuard tunnel between two Linux network namespaces acting as two VPN endpoints.

**Lab 25 — Port Scanning and Hardening · Slide 260**

**Objective:** Scan ports and reduce attack surface.

**Activity:** The learner uses nmap to enumerate exposed services on the host and disables the unnecessary ones.

**Lab 26 — Suricata IDS · Slide 270**

**Objective:** Detect intrusions with an ids.

**Activity:** The learner installs Suricata, updates its rule set, triggers a test signature, and inspects the resulting alert.

**Lab 27 — ACL with iptables · Slide 279**

**Objective:** Build a network acl.

**Activity:** The learner builds an iptables ACL that permits MySQL traffic from one subnet and denies it from everywhere else.

**Domain 5.0 — Network Troubleshooting (24%) · Slides 290–384**

Methodology · Layer 1/2 Diagnostics · Connectivity Tools · DNS/Routing/MTU · Packet Capture

**Lab 28 — Layer 1/2 Diagnostics · Slide 294**

**Objective:** Diagnose layer 1 and 2 faults.

**Activity:** The learner diagnoses physical and data-link faults using ip, ethtool and kernel interface counters.

**Lab 29 — ping, traceroute, mtr · Slide 303**

**Objective:** Locate loss and latency along a path.

**Activity:** The learner measures reachability, hop-by-hop path and continuous loss/latency using ping, traceroute and mtr.

**Lab 30 — DNS Troubleshooting · Slide 311**

**Objective:** Isolate name resolution problems.

**Activity:** The learner isolates DNS resolution problems using dig, +trace and resolvectl.

**Lab 31 — ARP and Neighbour Table · Slide 321**

**Objective:** Troubleshoot address-to-mac resolution.

**Activity:** The learner inspects and manipulates the ARP and IPv6 neighbour tables to resolve stale or duplicate entries.

### Lab 32 — Packet Capture (tcpdump/tshark) • Slide 330

**Objective:** Capture and analyse packets.

**Activity:** The learner captures DNS packets to a pcap and analyses them with tshark and the browser-based TIS PCAP Analyzer.

### Lab 33 — MTU and Path-MTU Discovery • Slide 341

**Objective:** Diagnose mtu and pmtud failures.

**Activity:** The learner tests interface MTU and simulates Path MTU Discovery using the Don't Fragment ping flag.

### Lab 34 — Routing Issues • Slide 349

**Objective:** Diagnose and fix routing problems.

**Activity:** The learner inspects the routing table, queries route selection and adds or removes static routes.

### Lab 35 — Network Emulation with tc • Slide 359

**Objective:** Emulate delay jitter loss and bandwidth.

**Activity:** The learner injects delay, jitter, loss and bandwidth limits on an interface with tc netem and tbf.

### Lab 36 — Service Reachability Triage • Slide 367

**Objective:** Triage service reachability by layer.

**Activity:** The learner isolates whether TCP, TLS, HTTP or the app is failing using curl, nc and openssl.

## Resources Required

- A computer with a modern web browser and internet access (per learner).
- The free Killercoda Ubuntu Playground (<https://killercoda.com/playgrounds/scenario/ubuntu>).
- Course slides and Learner Guide (downloadable from the LMS).
- For Lab 11 (Wireless): a device with a Wi-Fi radio and a Wi-Fi analyzer app.
- For Lab 32 (Packet Capture): the browser-based TIS PCAP Analyzer (<https://alfredang.github.io/pcapanalyzer/>).

## Lab Reference (aligned to N10-009 domains)

| Domain / Exam objective area       | Weighting | Labs                                                 |
|------------------------------------|-----------|------------------------------------------------------|
| Domain 1.0: Networking Concepts    | 23%       | Lab 1, Lab 2, Lab 3, Lab 4, Lab 5, Lab 6, Lab 7      |
| Domain 2.0: Network Implementation | 20%       | Lab 8, Lab 9, Lab 10, Lab 11, Lab 12, Lab 13, Lab 14 |

|                                            |     |                                                                        |
|--------------------------------------------|-----|------------------------------------------------------------------------|
| <b>Domain 3.0: Network Operations</b>      | 19% | Lab 15, Lab 16, Lab 17, Lab 18, Lab 19, Lab 20                         |
| <b>Domain 4.0: Network Security</b>        | 14% | Lab 21, Lab 22, Lab 23, Lab 24, Lab 25, Lab 26, Lab 27                 |
| <b>Domain 5.0: Network Troubleshooting</b> | 24% | Lab 28, Lab 29, Lab 30, Lab 31, Lab 32, Lab 33, Lab 34, Lab 35, Lab 36 |